

## Activity No.5

Date: / /

**Aim:** Identify Resin Identification Code (RIC) / Recycle Symbols of plastics on different plastic materials.

**Theory:** The Resin Identification Code (RIC) was developed in 1988 by the plastics industry association. It was created for workers in the plastic and recycling industry to be able to sort and recycle plastics more efficiently. Each RIC corresponds to a specific type of resin used in a plastic product.

Chronological order of when that plastic become recyclable :

- '1' signifies that the product is made out of **polyethylene terephthalate (PET)** (beverage bottles, cups, water bottles, soft drink, etc.)
- '2' signifies **high-density polyethylene (HDPE)** For example (shampoo bottles, water bottles, milk bottles and jugs, etc.)
- '3' signifies **polyvinyl chloride (PVC)** For example (PVC pipes, vinyl siding, flooring, window frames, etc.)
- '4' signifies **low-density polyethylene (LDPE)** For example (plastic bags, shopping bags, soft bottles, tubing, squeezable bottles, etc.)
- '5' signifies **polypropylene (PP)** For example (auto parts, industrial fibre, food containers, microwaveable wares, etc.)
- '6' signifies **polystyrene (PS)** For example (plastic utensils, CD cases, styrofoam, toys, hard packings, cafeteria trays etc.)
- '7' signifies other plastics, such as **acrylic, nylon, polycarbonate and polylactic acid (PLA)** for example. Baby feeding bottle, CD production etc.

\* As per **United State Army Garrison-Hawaii (USAG-HI)** and **United State Environmental Protection Agency (USEPA)** the plastic material having number 6, 7 or with no number are not recyclable.

**Apparatus:** Various types of plastic materials, like mineral water bottle, transparent cell phone's back cover, PVC pipe, soft drink bottles, plastic milk bags, etc.

**Procedure:**

1. Take one by one given plastic material.
2. Observe the bottom or lower side or all sides of plastic material and find the triangle  with number.
3. Mention your remark in observation table whether material is recyclable or not

**Observation and Result Table:**

| Sr. No. | Name of Plastic Materials                   | RIC Code | Remark (Recyclable or Not) |
|---------|---------------------------------------------|----------|----------------------------|
| 1       | Mineral Water bottle                        | 1        | Recyclable                 |
| 2       | Cell phone's transparent plastic back cover | 6        | not Recyclable             |
| 3       | PVC pipe                                    | 3        | Recyclable                 |
| 4       | Plastic beaker                              | 6        | not Recyclable             |
| 5       | Baby's feeding bottle                       | 7        | not Recyclable             |
| 6       | Soft drink bottle                           | 1        | Recyclable                 |
| 7       | Plastic container (empty)                   | 6        | not Recyclable             |

**Remark and sign of teacher:** .....

### MCQ

Select |✓| the most appropriate answer from given alternatives of each sub question.

1. The following symbol  indicate  
☒ a. universal recycling      b. universal decomposition  
c. universal reuse      d. use and throw
2. The RIC code on cafeteria tray is .....  
a. 1      b. 3      c. 4      ☒ d. 6
3. If RIC code on plastic material is 7 then which type of material will be there  
☒ a. polylactic acid      b. polyethylene terephthalate  
c. high density polyethylene      d. low density polyethylene
4. The RIC on auto vehicle part is .....  
a. 2      b. 4      ☒ c. 5      d. 7
5. The code used for recycle symbols on different plastic material is .....  
☒ a. RIC      b. MIC      c. TLC      d. PPC

### Short answer questions

1. What it indicates if there is no  such symbol on plastic material?

If there is no recycling symbol (RIC) on a plastic material, it indicates that the type of plastic cannot be easily identified for recycling purposes.

2. Write RIC for beverage bottles?

RIC = 1 (PET or PETE – Polyethylene Terephthalate)

3.  signifies which type of plastic material?

Polypropylene (PP)

4. Write RIC for food containers.

RIC = 5 (Polypropylene – PP)

5. What was the purpose for development of RIC in 1988?

The Resin Identification Code (RIC) was developed in 1988 to identify different types of plastics and to facilitate their sorting and recycling efficiently.

Remark and sign of teacher: .....